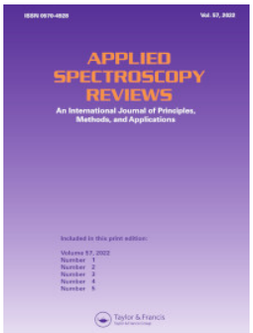

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REVIEW

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Direct through-container Raman spectroscopic analyses of samples housed in glass and plastic containers: a review

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ABSTRACT

This review summarizes current trends in direct through-container Raman spectroscopic analyses of housed samples without external sampling. Nonsampling Raman measurement is a relatively simple technique that avoids damaging or contaminating a sample's commercial value by opening the container. The most demanding aspect of such techniques is the need to minimize or eliminate spectral interference from the container when analyzing the housed samples. To reduce the spectral peaks associated with the container, which can be either fluorescent or nonfluorescent, various measurement configurations, including spatially offset Raman scattering and/or algorithm-based background subtractions, have been employed. Here, reports on the former technique are categorized according to their analytical purposes into qualitative chemical identification and quantitative determination of component concentrations of housed samples. Studies in each group are further classified by container material, either glass or polymer (plastic), as the optical properties of each, as well as the dominance of their Raman spectral peaks, are different. Prospects for advances and unresolved issues in through-container Raman measurement are also discussed.

KEYWORDS

Through-container Raman analysis; spatially offset Raman scattering; axially perpendicular offset scheme; glass container; plastic container

1. Introduction

Spectroscopic analysis of products housed in various containers, such as glass and plastic bottles, without direct sampling of the products themselves is a rapid and useful approach to sample identification and quality control, particularly in the security,[1,2] pharmaceutical,[3–6] and food sectors.[7–10] When through-container spectroscopic measurement is accomplished by illuminating an exterior laser source, the Raman peaks of both the sample and the container appear together, hampering recognition of the sample peak. Minimizing or eliminating container information without sacrificing sample information is, therefore, a crucial element of through-container spectroscopic analysis.[1–4,8,11] Spatially offset Raman scattering (SORS) is preferred for this task as flexible optical arrangements of laser-illumination and photon-detection make it possible to avoid or reduce interference from container photons (peaks).

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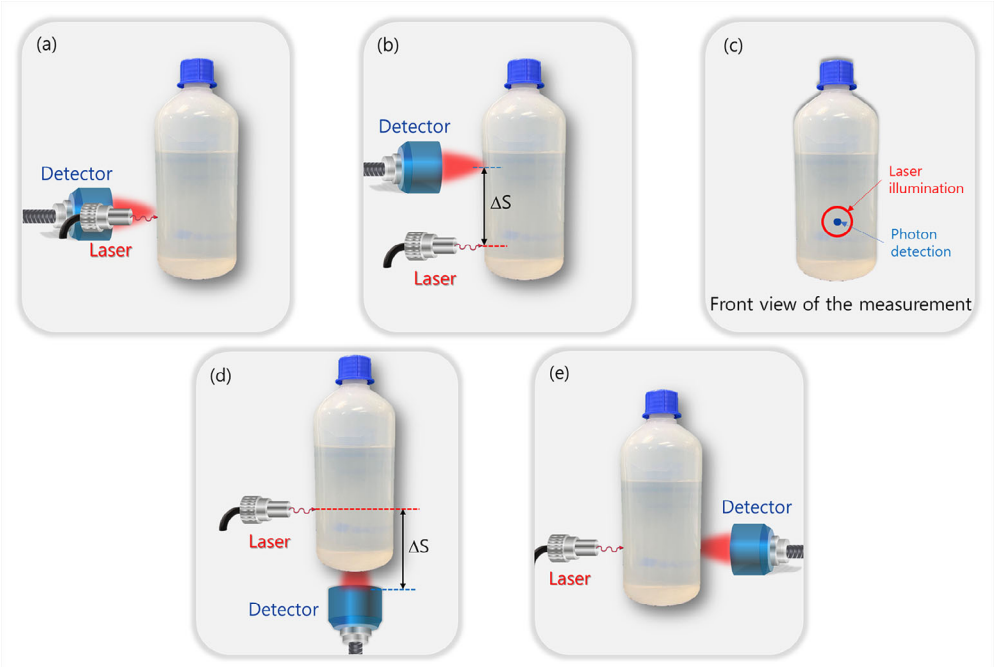

Figure 1. Diverse through-container measurement schemes: (a) back-scattering, (b) SORS, (c) inverse-SORS, (d) APO, (e) transmission.

Figure 1a provides a graphical description of a typical back-scattering Raman scheme. Because the laser illumination and photon detection points overlap, the obtained spectrum is dominated by container (wall) information. However, some photons from the laser propagate through the wall and interact with the housed sample, allowing for the collection of Raman information of the internal sample. As photons pass through a glass medium more easily than a plastic medium,[12,13] acquisition of information about an internal sample is easier when using a glass container compared with a plastic container.[2,8,14] While, the interference of co-presenting glass background is still a concern.

Figure 1b graphically describes a SORS scheme in which the Raman photon detection is separated (ΔS) from the point of laser illumination (the container wall). As a result, fewer photons generated from container arrive at the detector.[15] This is a basic strategy for reducing container information in collected spectra. However, the peak intensity of the housed sample also decreases due to larger attenuation of photons by longer travel to the detector. The optimal offset distance that can simultaneously maximize sample information and minimize container interference will depend on several parameters, such as the container material, container transparency, physical state of the contained sample (solid or liquid), and scattering or absorption properties of the sample.[16–18]

Modified SORS modalities have been developed to accommodate the analytical needs of specific studies. Among these modalities is an inverse-SORS geometry that collects Raman photons at the center of a sample with circular (ring-shaped) laser illumination around the center using optical fibers, as shown in Figure 1c. Inverse-SORS can improve the signal-to-noise ratio of the gathered spectrum due to its ability to collect a

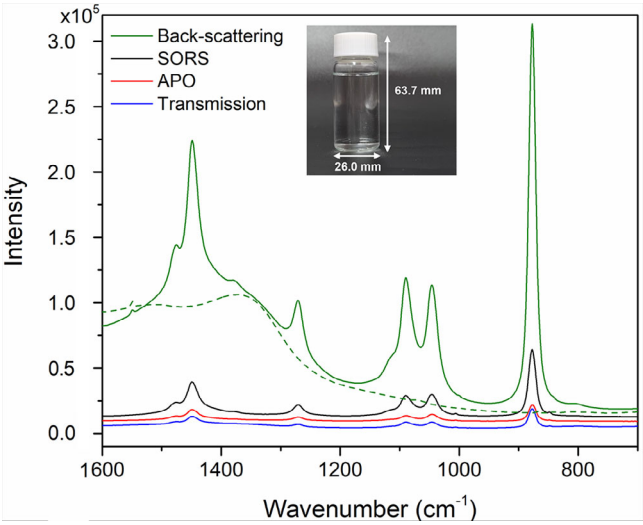

Figure 2. Raman spectra of 95% ethanol solution housed in a glass vial acquired using the back-scattering, SORS, APO, and transmission measurement schemes. The dashed-line spectrum is acquired from the empty glass vial.

greater number of photons along the ring illumination.[5] An axially perpendicular offset (APO) scheme, in which the laser illumination and detection of photons are arranged in an axially perpendicular geometry, as shown in Figure 1d, offers an alternative SORS modality. Container peaks can be minimized by adjusting the axial offset distance.[19] In a transmission Raman scheme, the laser illuminates one side of a sample-housed container and collects the photon that interacted with the sample by positioning the detector on the opposite side, as depicted in Figure 1e. However, this approach is not effective for large sample containers due to the attenuation of photons over the longer photon-propagation path.

Diverse through-container Raman measurements are available as described above. This review initially categorizes them into two groups according to the purpose of the analysis: qualitative chemical identification and quantitative determination of component concentration of housed samples. The studies in each group can be further classified by container material, glass and polymer (plastic), as the optical properties of each as well as the dominance of their Raman peaks in the final spectra differ.

2. Raman spectra of ethanol in glass and plastic containers collected with different measurement schemes

To examine the ability of reducing container spectral information, a 95% ethanol solution housed in a glass vial (diameter: 26.0 mm) was separately measured using the back-scattering, SORS, APO, and transmission schemes, as shown in Figure 2. The offset distance in the SORS and APO measurement was 6 and 20 mm, respectively. The picture of ethanol-housed glass vial is also presented in the figure. A laser (Invictus, Kaiser, wavelength: 785 nm, 400 mW) was illuminated on the wall of glass vial and generated Raman photons were collected using a PhAT probe (Kaiser). Each spectrum was

collected by accumulating 10 scans with a laser exposure time of 10 s. The spectra are arbitrarily offset for clear comparison. At a glance, the intensities of ethanol peaks in the back-scattering spectrum are predominantly larger than those in the remained spectra since a much larger amount of generated Raman photons can be detected in the back-scattering measurement. The peak intensities were smallest in the transmission measurement due to the farthest distance between the illumination and detection locations. Meanwhile, an underlying broad glass background in the 1800–900 cm^{-1} range appears together in the back-scattering spectrum (refer to the dashed-line spectrum of glass vial only). Although the peak intensities in the SORS, APO, and transmission spectra are considerably lowered, the glass background in each spectrum is effectively reduced as seen.

The same ethanol sample housed in a translucent polypropylene (PP) bottle was measured using the above four schemes (refer to the corresponding spectra in Figure 3b). The peak intensities are highest and lowest in the back-scattering and APO measurements, respectively, and the peaks of ethanol and PP are mixed (refer to the spectra of ethanol only and empty PP bottle in Figure 3a for comparison). To quantitatively examine the relative dominance of container information, the intensity ratio of ethanol peak at 877 cm^{-1} and PP peak at 803 cm^{-1} was calculated for the spectra in Figure 3b. The calculated ratios were 2.2, 4.3, 11.2, and 3.2, respectively, thereby informing that the APO measurement made the relative dominance of ethanol information over container information greatest. It is important to note that this comparison is for just demonstration and the suppression of container information will differ depending on diverse parameters, such as the offset distance, polymer type, container transparency, and physical state of the contained sample. The decreased peak intensity in the SORS, APO, and transmission measurements becomes a concerning issue when sensitivity is demanding for a targeting through-container Raman spectroscopic analysis.

3. Raman spectroscopic identification of samples in various containers

3.1. Identification of housed samples in glass containers

Table 1 provides a summary of previous studies involving the identification of housed samples in a variety of glass containers, with brief descriptions of the measured samples, container types, and spectral acquisition schemes, including the laser wavelengths. These studies were conducted to identify drugs (e.g., cocaine) dissolved in alcohol and ignitable liquids housed in clear and/or colored (brown, green, and blue) glass bottles, and various measurement schemes were used for the corresponding through-container analysis.

Back-scattering Raman measurements can recognize a housed sample in a glass container.[1,2,11] Although glass has a broad peak in the 1800–900 cm^{-1} range, its intensity is relatively low as glass is a material with weak Raman scattering. Because glass is optically transparent, a laser can easily pass through a glass wall to interact with the inner sample. Simple back-scattering is, therefore, a reasonable choice for through-glass container analysis, as long as the container does not generate fluorescence due to its color. As easily expected, background fluorescence obscures sample peaks and reduces identification accuracy.

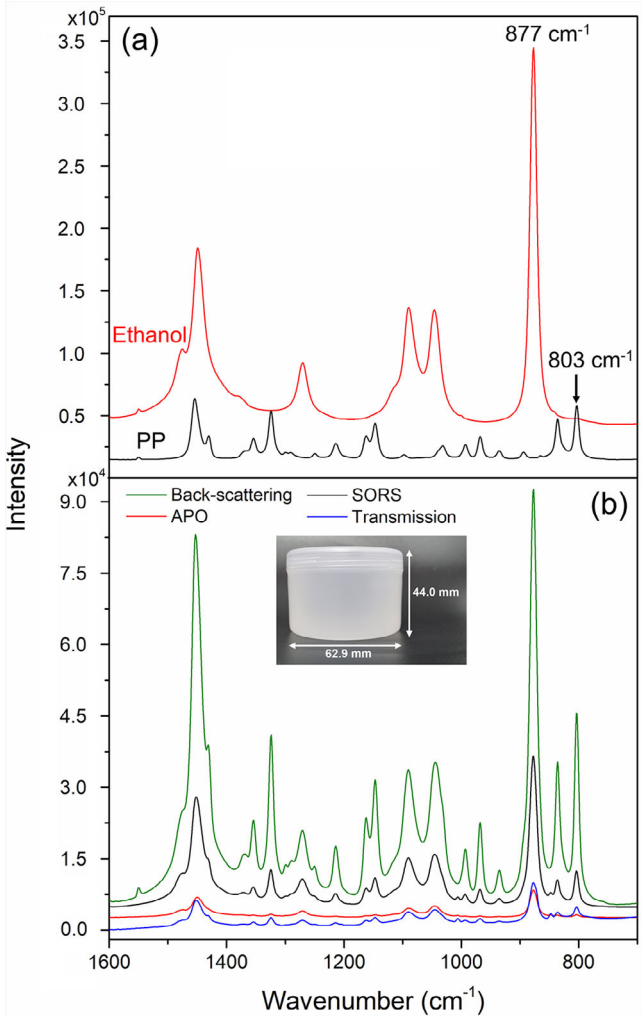

Figure 3. Raman spectra of ethanol only and empty PP bottle (a), and Raman spectra of 95% ethanol solution housed in the PP bottle acquired using the back-scattering, SORS, APO, and transmission measurement schemes (b).

Fluorescence is dependent on the color of the glass bottle as well as the wavelength of the laser excitation. For example, identification of samples housed in a green glass bottle, which is fluorescent upon irradiation with a 785 nm laser,[1,11] is challenging. Especially, when the concentration of a housed sample is low and fluorescence occurs, suppression or elimination of the fluorescence background are necessary to improve sample identification. A laser with a longer wavelength (such as 1064 nm) can decrease the chances of fluorescence[1] and diverse algorithms such as polynomial fitting are recommended to subtract the underlying background fluorescence.[2]

A SORS scheme can avoid detecting glass photons, which tend to gather around the container wall, and their signal can be distinguished from that of sample photons, which are widely spread in a container. Due to this ability, SORS,[7,8,20] APO,[14] and

Table 1. Summary of studies of Raman spectroscopic identification of housed samples in diverse glass containers.

Housed sample	Container	Measurement scheme	Note	Reference
Cocaine in ethanol	Clear, brown, light-green, dark-green glass bottles	Back-scattering (785 and 1064 nm)	Used 1064 nm excitation to reduce fluorescence from the dark-green glass bottle	[1]
Gasoline, sulfuric acid, diesel	Clear, blue, brown, green glass bottles	Back-scattering (785 nm) using a handheld spectrometer	Only green glass bottle hindered sample identification	[11]
Cocaine in alcohol beverage	Brown bottle	Back-scattering (785 nm)	Reduced the fluorescence background using a polynomial fitting	[2]
Hazardous liquids (CH_3COCH_3, CH_2Cl_2, etc.)	Commercial glass bottles (clear, amber, green)	Stand-off SORS (488 nm)	The distance between the laser illumination and photon collector was 6.7 m	[20]
Ethanol in spirit products	Green, brown, clear commercial glass bottles	SORS (830 nm) using a handheld spectrometer	Detected low-concentration ethanol without any contact with the sample	[7]
Ethylene glycol solution	Amber glass	SORS (830 nm)	Amber glass bottle produced fluorescence	[8]
Ethanol in water	Clear and brown glass bottles	APO scheme (785 nm)	The curved bottom of a glass vial generated a conical field of view for photon detection	[14]
Alcoholic sprits	Commercial sprit bottles	Inverse-SORS (785 nm)	The curved surface of a glass bottle acted as a cylindrical lens	[9]

inverse-SORS[9] schemes have been applied to the identification of samples housed in glass containers (Table 1). Because glass color, wall thickness, and the optical properties of the housed sample differed among the studies, optimal locations for laser illumination and photon detection that can minimize glass background interference without serious loss of sample information also varied. A proper offset distance was searched in each study to maximize identification performance. In some applications, the optical characteristics of the glass bottle itself effectively harmonized with the corresponding measurements. For example, in one APO scheme, the curved bottom of a glass vial generated a conical field of view for Raman photon detection, helping to separate the localized glass Raman photons at the vial wall from the widely distributed sample photons.[14] Similarly, a curved surface of glass bottle can act as a cylindrical lens in an inverse-SORS scheme.[9]

In general, Raman spectroscopic identification of housed samples in glass containers can be straightforward when the glass is not colored and the concentration of the housed sample is high. Suppression of the glass background becomes demanding when analyzing low-concentration samples. When some colored glass bottles generating fluorescence are used, fluorescence must be minimized to ensure more accurate sample identification. Various SORS schemes can help to minimize interference from fluorescence, and algorithms could effectively subtract the underlying fluorescence background.

3.2. Identification of housed samples in plastic containers

The Raman peak of glass lies between 1800 and 900 cm^{-1} range, while polymeric plastics such as polyethylene (PE), PP, polystyrene (PS), polyvinyl chloride (PVC), and

polyethylene terephthalate (PET) generate numerous and narrow Raman peaks over a wide spectral range. Because polymers are largely photon-diffusing materials, the distribution of generated container Raman photons is relatively broad, and separation of container and internal-sample photons in a measurement system can be challenging. The colors of plastic containers also tend to be more diverse, increasing the chances of fluorescence. Identification of housed samples in a plastic container, therefore, requires more thoughtful experimental design compared with glass through-container measurement.

Table 2 summarizes studies involving the identification of housed samples in a variety of plastic containers. In each study, either a housed sample or a container, or both, were fluorescent. Simple back-scattering schemes were employed using excitation of 1064 nm to permit deep penetration of laser photons as well as reduce fluorescence from the housed sample or colored plastic container, to measure fluorescent powder housed in a blister pack[3] and to analyze real alcohol housed in a purple-cream-colored plastic bottle.[1] Time-resolved Raman spectroscopy, which allows for temporal discrimination of photons from the surface of a sample and those from deeper layers, has been applied to noninvasive identification of explosives under opaque polymers with several millimeters of plastic packaging.[21]

Meanwhile, SORS schemes with various optical arrangements have been also applied to the effective separation of container photons against sample photons (Table 2). Acetaminophen, hydrogen peroxide (H_2O_2), and ammonium nitrate (NH_4NO_3) in colored containers have been identified using SORS measurements with specific optimal offset distances.[4,22,23] In addition, standard scaled subtraction of the two Raman spectra acquired at zero and nonzero spatial offsets has been used successfully to obtain a pure signature of the housed sample by canceling the spectral feature of an opaque PP container.[22] Minimization of background fluorescence using polynomial fitting and use of a smaller lens diameter helped to identify sugar in a white bottle and a colored sunlotion bottle.[23]

Several modified SORS configurations have also been examined. First, universal multiple angle Raman spectroscopy (UMARS) that can simultaneously acquire signals from a sample system at multiple angles to probe for any type of scattered sample photons at varying depths using optical fibers has been proposed. Five collection fibers were positioned around the periphery of a sample-containing high-density polyethylene (HDPE) container, at spacings of 60°. Ammonium nitrate and *t*-stilbene packed in the container were effectively recognized.[24] Second, an inverse-SORS scheme with an optimal radius (offset distance) of the ring-illumination zone produced distortion-free Raman spectra and a deeper penetration depth, enabling the identification of a paracetamol tablet in white PVC pipe (2 mm thick).[5] Third, a laser-defocused measurement scheme incorporating a wider object zone has been demonstrated. The scheme incorporated both illuminated and nonilluminated zones, which allowed for selective subsurface sampling beneath the container, so it successfully recognized paracetamol housed in a white plastic bottle.[6] Fourth, a time-resolved (TR) measurement was combined with SORS (TR-SORS) to take advantage of the temporal and spatial isolation abilities of container photons. The use of a time-gated intensified charged coupled device effectively rejected the container background, enhancing the identification of explosive precursors housed in a fluorescing plastic packaging.[25,26] Fifth, the ability of a SORS-based dual-offset optical

Table 2. Summary of studies of Raman spectroscopic identification of housed samples in diverse plastic containers.

Housed sample	Container	Measurement scheme	Note	Reference
Antimalarial artesunate tablet	Blister packs (material not specified)	Back-scattering (785 and 1064 nm) using handheld spectrometer	Used 1064 nm excitation to diminish fluorescence background	[3]
Cocaine in ethanol	Colored plastic (material not specified)	Back-scattering (1064 nm)	1064 nm excitation was effective to reduce background fluorescence	[1]
DNT, ethyl centralite, and diphenylamine (explosives)	White opaque PVC, POM, PS, PE, and PTFE containers	Time-gated back-scattering (460 nm)	Subtraction from the spectrum with no delay time to the spectrum with optimum delay time	[21]
Paracetamol and ibuprofen capsules	Translucent blister packs (material not specified)	SORS (827 nm)	The green/yellow capsules generated fluorescence even in the SORS measurement	[4]
H_2O_2 solution and moisturizer cream	White and translucent plastic (material not specified)	Adapted SORS (830 nm)	Regulated the angle of laser illumination for maximal suppression of fluorescence	[22]
Sucrose powder	White plastic (material not specified)	SORS (830 nm)	High-order polynomial fitting to subtract the fluorescent background	[23]
NH_4NO_3, DCB, and *t*-stilbene powders	White HDPE	UMARS scheme (830 nm)	Searched an optimal angle able to maximize the spectral features of contained samples	[24]
Paracetamol	White PVC	Inverse-SORS (827 nm)	Allowed higher sensitivity and deeper penetration depth compared to those of SORS scheme	[5]
Paracetamol	White plastic (material not specified)	Defocusing Raman (830 nm)	The defocused concept scheme was simple, but less effective in decreasing container information	[6]
NH_4NO_3, $BaSO_4$, and so on	HDPE of different colors	Time-resolved SORS (785 nm)	Achieved the high selectivity, while the low signal-to-noise ratio was a concerning issue	[25]
NH_4NO_3	Yellow PS	Time-resolved SORS (785 nm)	The signal-to-noise ratio decreased with lengthening the detector gate delay time	[26]
PMMA, trisodium citrate, glucose, and so on	PE, PP, PS	Dual-offset Optical Probe (785 nm)	Three bundles of optical fibers were used for excitation with 2 different offset distances	[27]
Ethanol solution	Transparent PET and blue PP	Handheld SORS (830 nm)	High-order polynomial fitting to subtract the background fluorescence	[10]
NH_4NO_3, H_2O_2, and $NaClO_4$	HDPE and fabric garment	Portable inverse-SORS (785 nm)	The identifications were executed under normal indoor and outdoor illumination conditions	[28]
$NaClO_3$ and isopropyl alcohol	White HDPE and opaque LDPE	Stand-off SORS at 12 m distance (532 nm)	Combined stand-off and SORS measurements. Fluorescence reduced by half with 47 mm offset	[29]

probe system (DOOP) to identify 20 different chemicals concealed in various containers has been studied. The DOOP was designed to eliminate the container background peak by subtracting two spectra obtained at two different offsets (5 and 10 mm), which is equivalent to subtracting the pure container background feature from the spectrum containing information from both the container and the inner sample.[27]

To apply SORS to practical at-site measurement, several explorations have been conducted. First, when a handheld spectrometer was used to identify an internal ethanol solution by varying the plastic thicknesses up to 21 mm, the ethanol signals were only observable at depths below 9 mm.[10] Multivariate analysis was necessary for the identification, particularly when the spectral features of the container were similar to those of the housed samples. Second, the utility of a portable inverse-SORS system was evaluated for the identification of explosive precursors and drugs in plastic packaging. The measurements were taken in daily-life conditions, such as under incandescent or fluorescent lighting and daylight.[28] Third, to demonstrate remote-detection capability, a stand-off SORS scheme capable of collecting Raman signals of samples at a distance of 12 m using a 6-inch telescope was studied. Using this scheme, Raman spectra of sodium chlorate ($NaClO_3$) powder and isopropyl alcohol housed in an opaque PE bottle (1.5 mm thick) were successfully collected at the remote distance.[29]

As with through-glass container measurement, housed samples were more easily identified when the plastic containers did not exhibit florescence, and more dedicated spectral acquisition schemes were necessary to improve identification performance when using fluorescence-generating plastic containers. Time-gated separation of fluorescence would be more intrinsic and is promising; however, high instrumentation cost and the low signal-to-noise ratio of the acquired spectrum limit further use of such systems. Identification of an internal sample becomes difficult as the thickness of the plastic wall increases, although this may not be a critical limitation in measurements taken through glass containers, and SORS measurements using portable instrumentation has drawn increasing attention due to demands for practical at-site sample test.

4. Raman spectroscopic quantitative analysis of samples in various containers

4.1. Quantitative analysis of samples housed in glass containers

Beyond the identification of housed samples in a glass container, the direct determination of a housed sample's concentration is useful, particularly for simple quality control and assurance of many commercial packed products. Diverse research efforts have been devoted to this subject, as summarized in Table 3. In these reports, either nonoverlapping sample peaks with the glass background were utilized for quantitative analysis or the glass background was minimized using spectral pretreatment methods to highlight the sample peak. Again, either back-scattering or SORS measurement schemes were typically chosen, depending on the analytical needs of each application.

Back-scattering based on wide-area illumination (WAI) was used to determine the acetaminophen concentration of a turbid pharmaceutical suspension housed in a glass vial. The reproducibility of WAI-based measurement was superior to that produced with a conventional small-spot scheme due to the more representative spectroscopic

Table 3. Summary of studies of Raman spectroscopic quantification of housed samples in diverse glass containers.

Housed sample	Container	Measurement scheme	Note	Reference
Acetaminophen suspension in sucrose medium	Glass vial	WAI back-scattering (785 nm)	Use of the spectral range avoiding the glass vial background provided most accurate analysis	[30]
Antineoplastic drugs in sodium chloride solution	Glass vial	Microscope back-scattering (633 nm)	Careful examination of container characteristics was necessary to set up an optimal measurement	[31]
Mixtures of cocaine HCl powder and tablet	Glass vial	Handheld back-scattering (785 nm)	Brown powders and colored tablets showed fluorescence, thereby degrading the accuracy	[32]
Taxane drugs in NaCl solution	Glass vial	Handheld back-scattering (785 nm)	Various spectral processing methods were examined to improve the accuracy	[33]
Whiskeys	Clear bottles, and transparent and green vials	Handheld back-scattering (785 and 1064 nm)	For the green bottle, 1064 nm excitation reduced fluorescence as well as sample information	[34]
Doxorubicin and epirubicin in NaCl solution	Hospital-adopted glass vials	Handheld back-scattering (785 nm)	Regulated focal distance with a proper spectral processing method to limit the glass background	[35]
Liquid injectables (doxofylline, levofloxacin)	Ampuoles and glass bottles	Portable back-scattering (785 nm)	Used the loading spectra to separate the glass signals and extracted sample signal using CLS	[36]
Methanol in whiskey and vodka	Commercial colored glass bottles	Handheld back-scattering (1064 nm)	The result was poor when the gin in a dark brown bottle was measured due to the light absorption	[37]
Mixture of extra virgin olive oils with three adulterants	Clear glass bottle	Handheld SORS (830 nm)	Diverse chemometric analyses were examined to overcome the overlapping glass peak	[38]
Gemcitabine injection powder samples	Medical glass vial	ASI scheme (785 nm)	The decreased glass peak by lengthening the distance between laser illumination and detector	[39]

sampling. The use of a narrower spectral range, excluding the glass peak, for partial least squares (PLS) resulted in more-accurate concentration measurements.[30] The concentrations of five different antineoplastic drugs (5-fluorouracil, gemcitabine, cyclophosphamide, ifosfamide, and doxorubicin in aqueous 0.9% sodium chloride) in glass vials were also measured using a back-scattering scheme. The concentrations of the housed drugs were low. When the whole spectral range was used for PLS, the prediction accuracies were acceptable for each drug but 5-fluorouracil. To improve the prediction of 5-fluorouracil concentration, separate PLS models for low- and high-concentration ranges were ultimately adopted.[31]

To demonstrate the capability for at-site measurement, several research groups executed through-glass container measurements using a handheld spectrometer (back-scattering mode), as described in six reports. Two of the groups quantitatively analyzed binary cocaine mixtures (0–100 wt%) with procaine, paracetamol, or caffeine[32] and three different taxane drugs (cabazitaxel, docetaxel, and paclitaxel) in laboratory glass

vials.[33] When brown-colored drugs such as heroin and methylenedioxymethamphetamine (MDMA) were analyzed, the obtained limits of detection (LODs) were approximately 30–40 wt%.[32] To suppress the signal from the glass and the noninformative spectral background, Savitzky-Golay derivatives and standard normal variate (SNV) were utilized as spectral preprocessing tools.[33]

The other four groups dealt with measurements of samples housed in real commercial glass containers. The age and alcohol content (40–60% by volume) of market-available Scotch whiskeys were measured. However, when the samples were contained in green glass bottles, the fluorescence was severe, and the corresponding samples were transferred into transparent vials for analysis.[34] Concentrations of doxorubicin (0.1–2 mg/mL) and epirubicin (0.08–2 mg/mL), which are isomeric drugs used in cancer treatments, separately housed in medical glass vials were determined.[35] Similarly, liquid injectable doxofylline ampoules and levofloxacin lactate in large glass bottles were measured. The resulting accuracies were acceptable; however, the low sensitivity of portable (handheld) Raman measurement was reportedly a concern.[36] The use of chemometric methods was, therefore, necessary to minimize interference from different glass containers and allow for extraction of weak sample signals.[35,36] Using excitation of 1064 nm, the quantification of methanol (a substitute alcohol for ethanol) in counterfeited spirit drinks was demonstrated. The LODs based on the measurements of four different spirit types ranged from 0.23% to 0.39%, lower than the general EU limit (0.5%). The 1064 nm excitation clearly reduced the fluorescence contributions in the acquired sample spectra.[37]

Handheld spectrometers improve the ease of taking SORS measurements. Back-scattering and SORS measurements were performed separately using a handheld spectrometer to determine the concentration of extra virgin olive oil (0–100%) mixed with three adulterants. The accuracy obtained with back-scattering was unsatisfactory, while the SORS measurement was able to construct a robust PLS model due to its deep-sampling ability.[38] In another approach to SORS measurements, an axially slanted illumination (ASI) scheme used a slanted irradiating laser on the headspace side of a glass container and a detector positioned beneath the container. This scheme was to increase the distance between the laser illumination point (glass wall) and detector location to minimize the number of glass photons reaching the detector. Using ASI, the concentration of gemcitabine (42.9–58.2 wt%) in gemcitabine injection powder was accurately measured due to the enhanced gemcitabine spectral feature by the diminished glass background.[39]

4.2. Quantitative analysis of samples housed in plastic containers

Table 4 lists reports on direct quantitative analyses of housed samples in various plastic containers. Typical back-scattering measurements were employed in some of these studies. Determination of paracetamol concentrations in ternary mixtures inside PVC blister packages was performed and multivariate curve resolution-alternating least squares (MCR-ALS) analysis was used to minimize spectral interference from blister-pack materials.[40] A back-scattering WAI scheme was used to determine the hydrogen peroxide concentration (1–5 wt%) in antiseptic solutions in red opaque HDPE bottles[41] and

Table 4. Summary of studies of Raman spectroscopic quantification of housed samples in diverse plastic containers.

Housed sample	Container	Measurement scheme	Note	Reference
Ternary pharmaceutical mixtures (acetaminophen)	White PVC blister	Microscope back-scattering (785 nm)	Tried to build a calibration model using a multiset structure with samples in and out of the blister	[40]
H_2O_2 solution	Red opaque HDPE	WAI back-scattering (785 nm)	The prediction accuracy maintained even with the *x-y-z* variation of container positions	[41]
Urea cream	LDPE bottle	WAI back-scattering (785 nm)	Used the spectral range for quantification at which the urea peak was most dominant	[42]
Gemcitabine in NaCl solution	Transparent PP and PVC injection bags	Handheld back-scattering (785 nm)	Demonstrated ability for at-site quality control of finished medical products	[43]
Hand sanitizer	HDPE, PET, and PP bottles with different transparences	Portable SORS (830 nm)	Using chemometrics, the alcohol content was well predicted regardless of the container types	[44]
Drug mixtures (API: acetaminophen or other)	White opaque PP	Inverse-SORS (785 nm)	The larger sampling volume compensated for sample inhomogeneity, yielding the low prediction error	[45]
Urea gel	White opaque PP	APO scheme (785 nm)	Narrowing detection window to escape container photons, leading to the improvement of accuracy	[46]
Ethanol gel	Pink translucent PP (oval shape)	APO scheme (785 nm)	Accuracy was maintained although the oval container was rotated at each measurement	[19]

urea concentration (7.2–12.8 wt%) in cream formulations in LDPE bottles.[42] When measuring hydrogen peroxide concentrations, prediction accuracy was not degraded even with the *x-y-z* positional variation (±5 mm) of the bottle,[41] and the correct sample representation of cream samples by the WAI scheme was attributed to reliable through-container quantitative analysis.[42] Using a handheld spectrometer, the quantification of gemcitabine intravenous drugs contained in chemotherapy plastic bags (multilayer PP primary bag film) was demonstrated. The housed samples were the commercial formulations of gemcitabine (40 mg/mL), with disodium phosphate, ethyl alcohol, hydrochloric acid, and sodium chloride as excipients. Among 1628 gemcitabine plastic bag samples, 97.3% were controlled by the proposed method.[43]

Several SORS schemes have been used for quantitative analysis through plastic containers. For example, the alcohol content of hand sanitizers in plastic containers made of HDPE, PET, and PP was measured. Support vector regression allowed sustainable prediction performance (R^2 greater than 0.98) despite varying transparencies and container sizes.[44] Modified SORS schemes such as inverse-SORS and APO were employed for the same purpose. The potential of an inverse-SORS scheme was demonstrated by measuring two sets of mixtures containing acetaminophen and phenylephrine hydrochloride, which were strong and weak Raman scatterers, respectively, in an opaque

white PP container.[45] Using an APO scheme, the urea concentration (7.5–12.5%) of a urea gel in a PP container was measured directly. To minimize the intensity of the peaks associated with the containers, and make the generated container Raman photons less detectable to the bottom-positioned detector, the detection window was narrowed.[46] The APO scheme proved to be a versatile tool for reproducible measurement of the housed sample (ethanol gel) in a noncircular (oval) container even under change of container orientations. In APO scheme, Raman photons of the sample traversed the bottom of the container even with the orientation change, making Raman sampling volumes for the internal sample consistent, and subsequent spectra were reproducible. However, in the cases of either back-scattering or transmission measurements under the same circumstances, the Raman sampling volumes varied, lowering the reproducibility of the sample peak intensities and degrading the accuracy.[19]

5. Conclusions and future prospects

Direct through-container Raman spectroscopic analysis of samples housed in glass or plastic containers is attractive to analytical chemists as it enables sampling-free, simple, and rapid analysis. It is an ideal analytical tool for noninvasive quality control and quality assurance of housed samples, such as those used in the pharmaceutical, cosmetics, and food sectors. Back-scattering measurements are relatively straightforward but offer an acceptable approach only when the container is nonfluorescent and or has a thin wall. When either a housed sample or a container is fluorescent, and/or the container wall is thick, SORS, or modified SORS schemes more effectively suppress the fluorescence and/or container spectral feature.

Because demand for at-site measurement is rising along with the need for rapid field quality control and forensic analysis, the use of portable Raman instrumentation for target analysis is expected to increase. Low signal-to-noise ratios and the inferior reproducibility of portable Raman measurement has proven to be key issues to overcome for application of this approach. Considerable research and development efforts by instrument manufacturers will be required to address these drawbacks. Time-gated Raman measurement offers considerable potential for effective through-container analysis, while the low signal-to-noise ratio of the collected spectra and high instrumentation costs must be addressed.

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